H2 ANDERSON JUNIOR COLLEGE HIGHER 2

2018 JC2 PRELIMINARY EXAMINATIONS

CANDIDATE NAME	

PDG					INDEX NUMBER			

H2 BIOLOGY 9744/02

Paper 2 Structured Questions

11 SEPT 2018
TUESDAY

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name and PD group on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graph
Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** questions in the space provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	/9
2	/9
3	/11
4	/11
5	/11
6	/10
7	/11
8	/9
9	/11
10	/8
	/100

Answer **all** questions

1 Fig. 1.1 shows the effect of pH on the activity of a protease enzyme at the optimal temperature of 37°C.

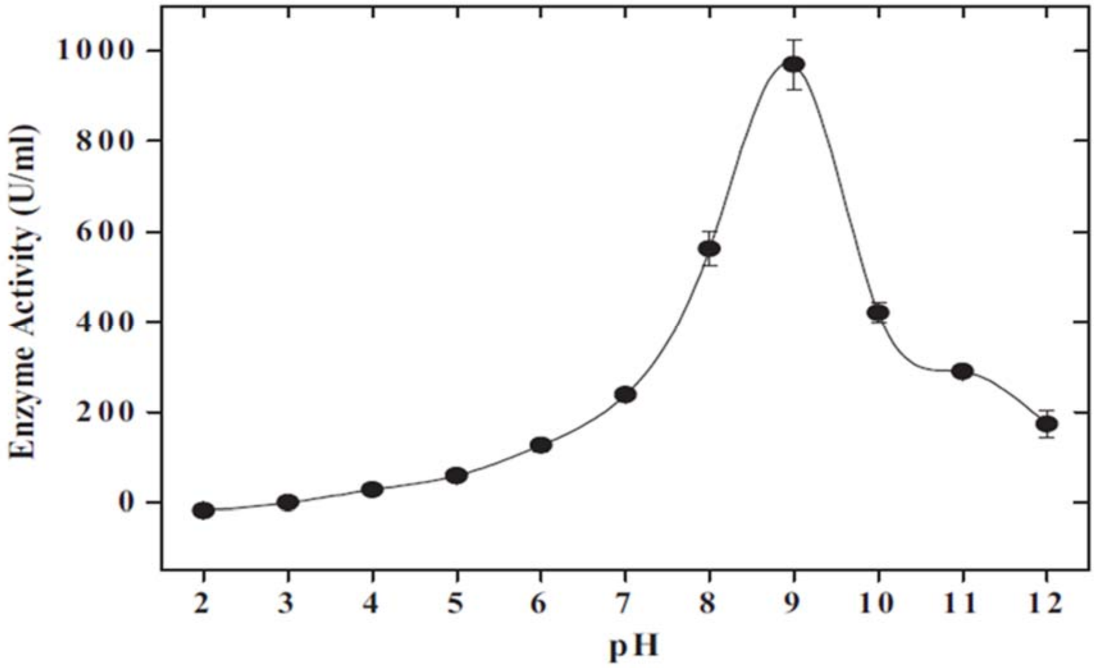

Fig. 1.1

(a) Draw, on Fig. 1.1, the approximate shape of the curve if the same experiment is conducted at 25°C. [1]

(b) Explain with reasons the shape of the curve you have drawn.

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(c) Using information from the graph, explain why proteases stored in vesicles with pH 7.2 cannot break down vesicular membrane proteins and suggest how these proteases can be activated through increase in pH.

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[Total: 9]

2 RNA molecules play important roles within cells. One of the major types of RNA found in all cells is the ribosomal RNA (rRNA).

Fig. 2.1 shows rRNA molecules forming the large ribosomal subunit in eukaryotes.

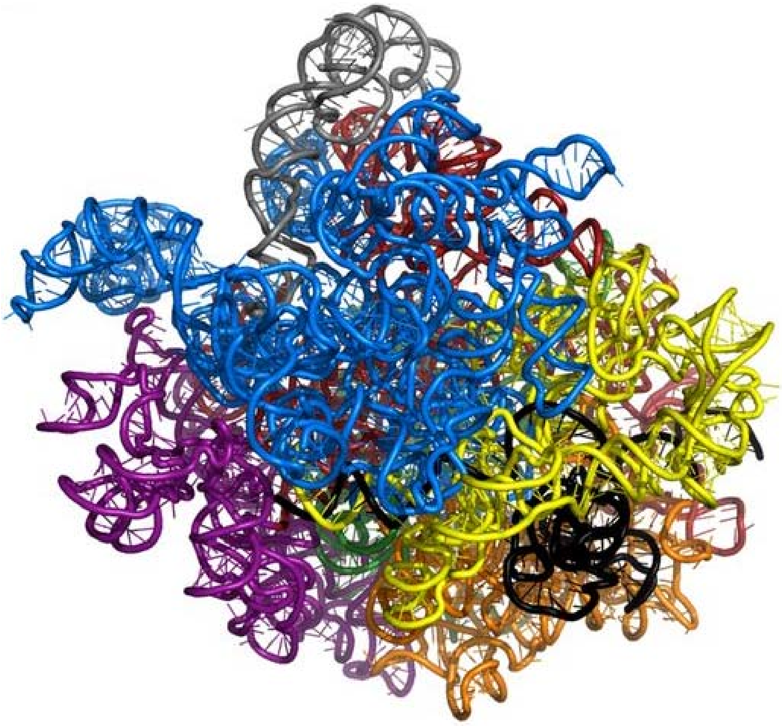

Fig. 2.1

(a) Explain why the rRNA molecules must adopt the shapes shown in Fig. 2.1.

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.... [3]

(b) Another important RNA molecule found in eukaryotic cells is the telomerase RNA. Telomerase RNA is found within the telomerase enzyme, an enzyme essential for elongating telomeres.

(i) Outline how RNA molecules such as telomerase RNA and rRNA are synthesised.

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Fig. 2.2 shows the mode of action of telomerase.

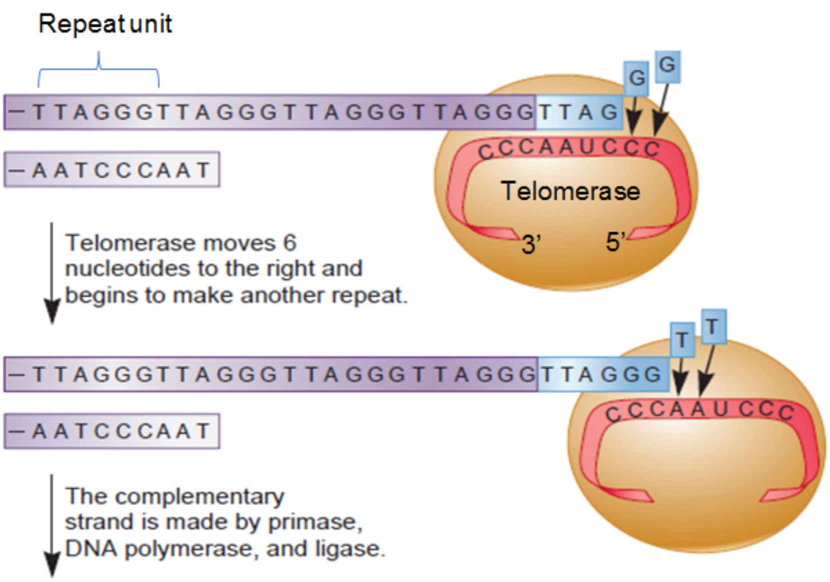

Fig. 2.2

(ii) Describe **three** visible differences between telomere elongation shown in Fig. 2.2 and translation.

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.... [3]

[Total:9]

3 Eukaryotes regulate the expression of their genes at various levels of protein synthesis.

(a) Describe the effect of histone acetylation on gene expression.

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(b) Eukaryotic gene expression can also be regulated at translation initiation after the mRNA is synthesised.

Fig. 3.1 shows translation occurring on a eukaryotic mRNA.

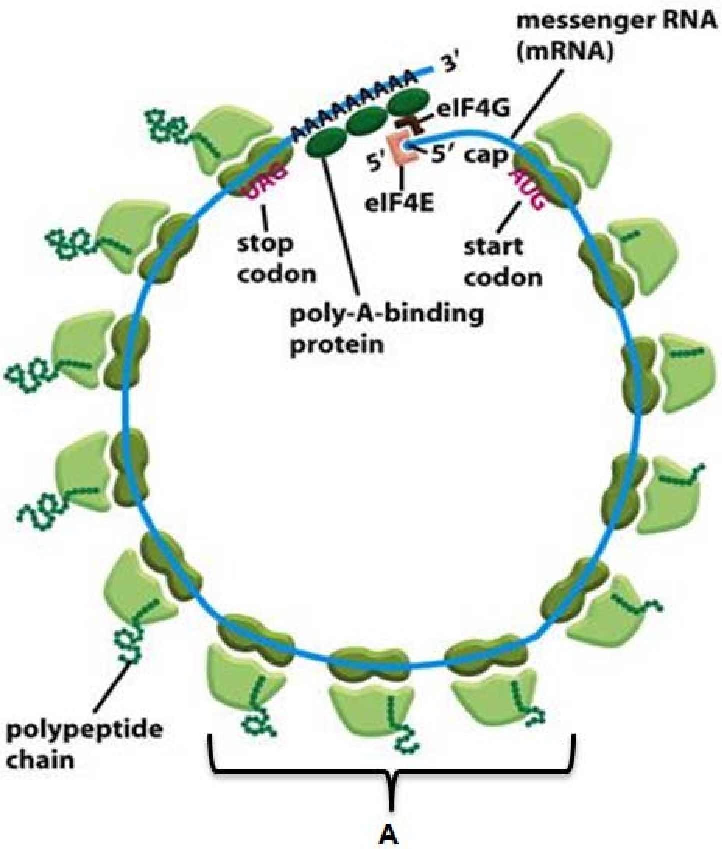

Fig. 3.1

(i) Explain the significance of the pattern of translation, labelled **A** in Fig. 3.1**.**

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.... [2]

(ii) During translation initiation, translation initiation factors like eIF4E and eIF4G form part of a complex which aid in recruiting ribosomal subunits to mRNA.

With reference to Fig. 3.1, describe the role of the poly-A tail and 5' cap in the assembly of ribosomes.

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.... [2]

(iii) State **one** other function of the poly-A tail.

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.... [1]

(c) In mammals, sex is determined by the X and Y chromosomes, females being XX and males XY. In females, the expression of all the genes on one of the two X chromosomes in each cell is inactivated throughout the life of the female. This ensures that the effective dosages of products of X-linked genes are equal in males and females since a double dose of X-linked genes may potentially be toxic.

Suggest if the inactivation of gene expression on the X chromosome occurs via chromatin modification or at translation initiation. Explain your answer.

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[Total: 11]

4 Viruses infect both eukaryotic and prokaryotic cells.

Fig. 4.1 is an electron micrograph of a eukaryotic cell infected by an enveloped virus such as the Zika virus.

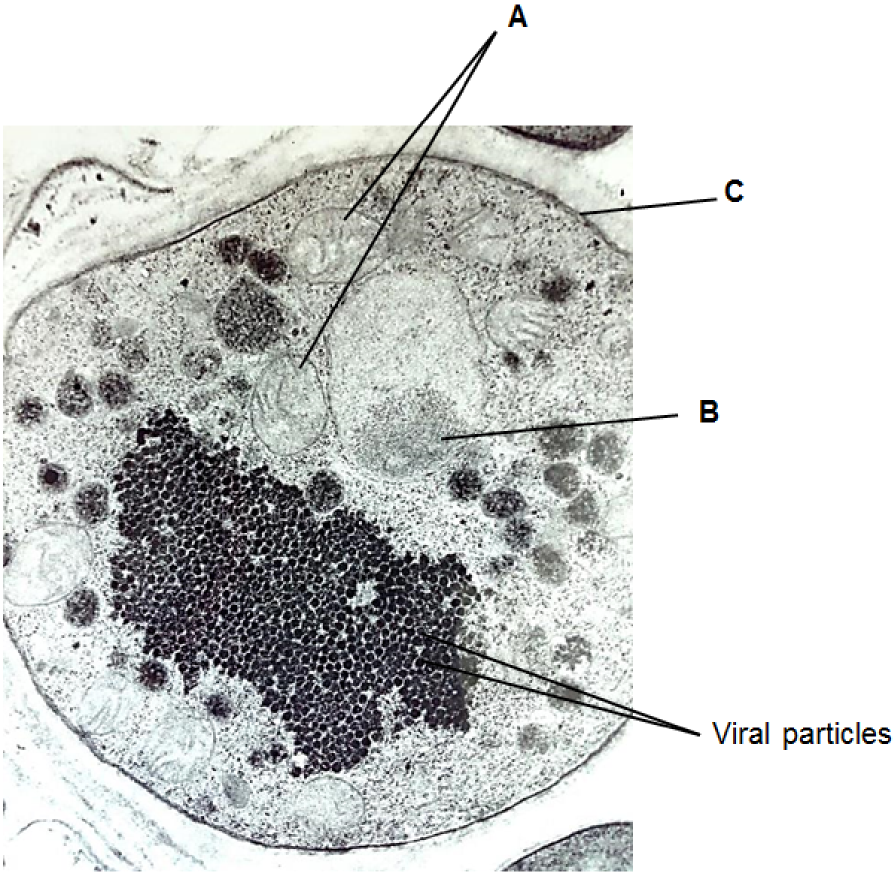

Fig. 4.1

(a) **(i)** Identify structures **A** and **B** visible in Fig. 4.1.

A: ....

B: .... [2]

(ii) Describe the role of structures **A** and **C** in the reproductive cycle of viruses such as the Zika virus.

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.... [3]

Bacteriophages are viruses that infect prokaryotic cells. The lambda bacteriophage is an example of a temperate bacteriophage that infects *Escherichia coli.*

Fig. 4.2 shows the changes in the number of extracellular lambda phages after they are introduced into a culture of *E.coli.*

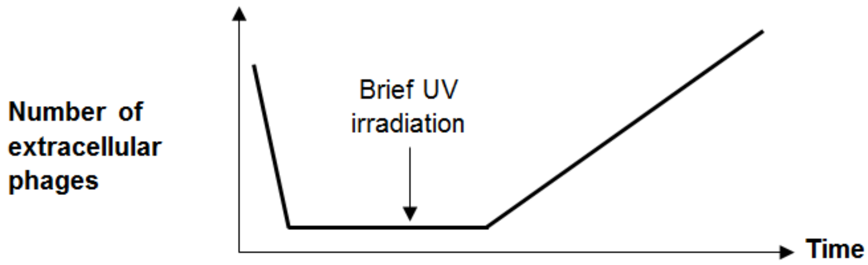

Fig. 4.2

(b) With reference to the reproductive cycle of temperate bacteriophages,

(i) explain why the number of phages increased after a brief UV irradiation.

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(ii) explain why bacteriophage infection may be beneficial to bacteria population.

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.... [3]

[Total:11]

5 Skin cancer cells may be grown in culture and examined using the technique of immunofluorescence in which antibodies are used to attach fluorescent dyes to specific molecules within the cells.

Fig. 5.1 is an immunofluorescent light micrograph of skin cancer cells. A particular type of protein is stained with the dye and appears as pale regions in the skin cancer cells.

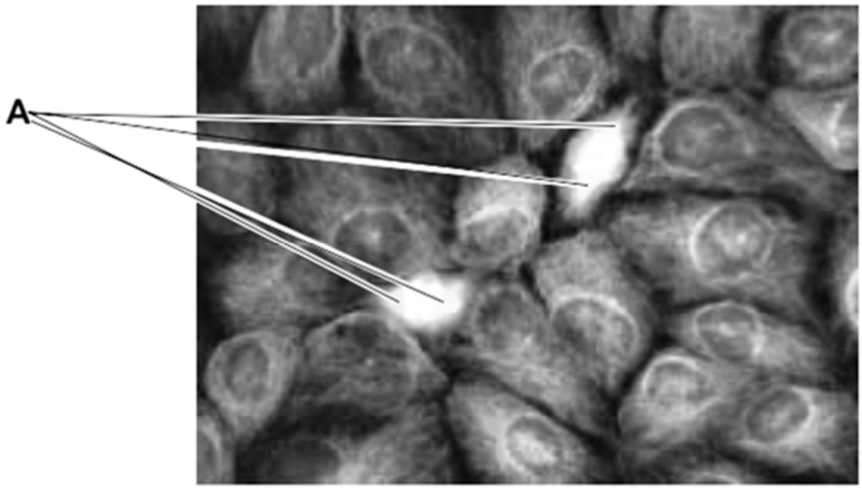

Fig. 5.1

(a) **(i)** Before the skin cancer cells could be stained with antibodies, the cells had to be fixed and treated with a mild detergent to increase the permeability of the cell surface membranes.

Explain the purpose of this step.

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.... [2]

(ii) There are two cells in the process of dividing. Each of these cells has two areas stained heavily, labelled **A** on Fig. 5.1.

Suggest the identity of these two areas and outline their functions in these cells.

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.... [3]

(iii) Suggest why the proteins stained in the cytoplasm of the non-dividing cells in Fig. 5.1 are not evenly distributed.

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.... [1]

(b) Explain **two** ways in which the behaviour of chromosomes in prophase of meiosis I differ from prophase of mitosis.

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.... [2]

(c) Some chemicals known to inhibit the cell cycle are used as drugs for the treatment of cancer.

A particular drug was found to be most effective when applied to cancer cells in the G2 phase of the cell cycle.

Suggest the possible mechanism of this drug.

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.... [3]

[Total: 11]

6 A castor oil plant, taken from a line which was known to be pure-breeding for the black and smooth seed coat, was crossed with a plant of unknown genotype.

The cross gave the following F1 results.

black, smooth seed	1315
red, wrinkled seed	1370
red, smooth seed	21
black, wrinkled seed	19

(a) Draw a genetic diagram to explain these results.

Use the following symbols.

R red seed; **r** black seed; **N** wrinkled seed; **n** smooth seed.

[4]

(b) In another experiment involving two other characteristics of the castor plant, repeated test crosses with a plant which is heterozygous at both gene loci only produce progeny with 2 phenotypic classes instead of the expected 4.

Explain the above observation.

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(c) In a different plant species, the type of seed coat is controlled by a single gene, B/b.

This gene encodes for an enzyme that converts the glucose formed during photosynthesis into starch for storage. Starchy seeds remain relatively smooth.

On the other hand, the homozygous recessive condition disrupts this conversion, producing seeds with high sugar content. When these seeds dry up, they shrivel and become wrinkled.

Explain why heterozygous plants for this gene, Bb, have the same phenotype as homozygous dominant plants, BB.

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[Total:10]

7
(a) Explain the part played by water in the production of ATP during photosynthesis.

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Most ATP is made in cells by membrane systems that create proton gradients by pumping protons from one compartment to another.

Fig. 7.1 shows three such membrane systems.

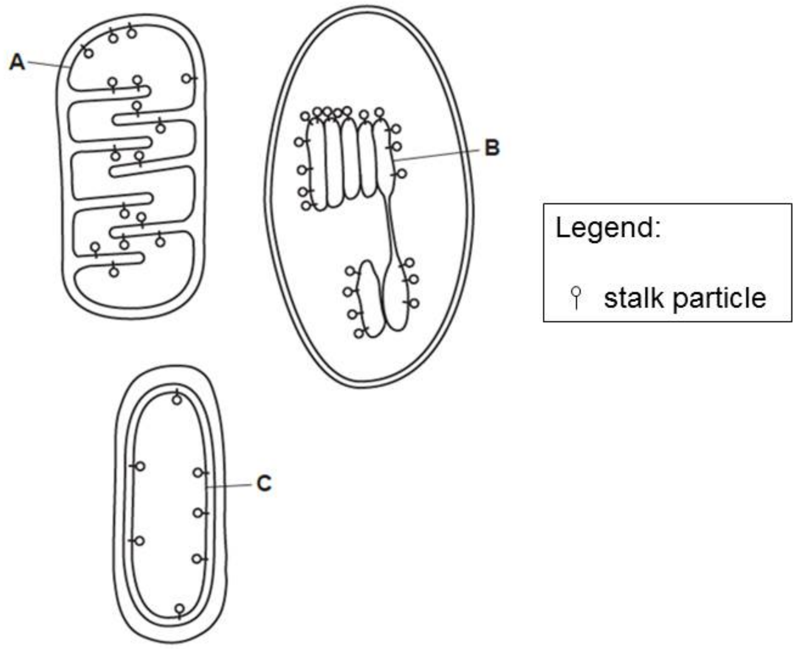

Fig. 7.1

(b) **(i)** Draw arrows onto each of the membrane systems in Fig. 7.1 to show the direction in which protons are pumped. [1]

(ii) Describe the role of membrane **B**.

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Fig. 7.2 summarises the reactions which occur in the Calvin cycle.

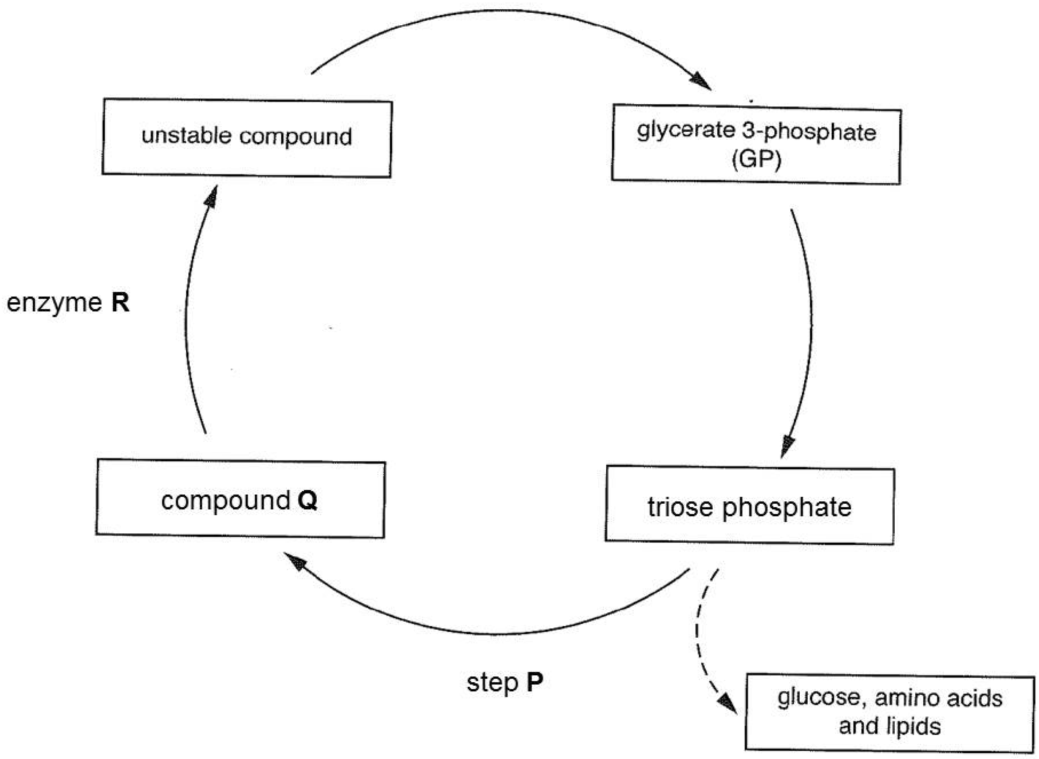

Fig. 7.2

(c) **(i)** Describe step **P**.

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.... [2]

(ii) Some biologists describe enzyme **R** as 'the most important enzyme in our biosphere'.

Explain why they might hold this opinion.

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.... [3]

[Total:11]

8 Reproduction in seahorses, *Hippocampus*, is unusual as it is the male rather than the female that becomes pregnant. The male has a brood pouch located on its tail. The larger the male, the larger the pouch. The female transfers unfertilised eggs into the pouch. The larger the female the more eggs are produced that can be transferred to the brood pouch. The male releases sperm onto the eggs and they are fertilised. The male carries the developing brood for a period of several weeks until he finally gives birth.

Research into seahorse populations has revealed the following.

- They are monogamous. A male and female remain together for the whole mating season.
- Within a population, mates are selected by size. Large females mate with large males and small females mate with small males.
- Few intermediate sized individuals are produced and they have low survival rate.

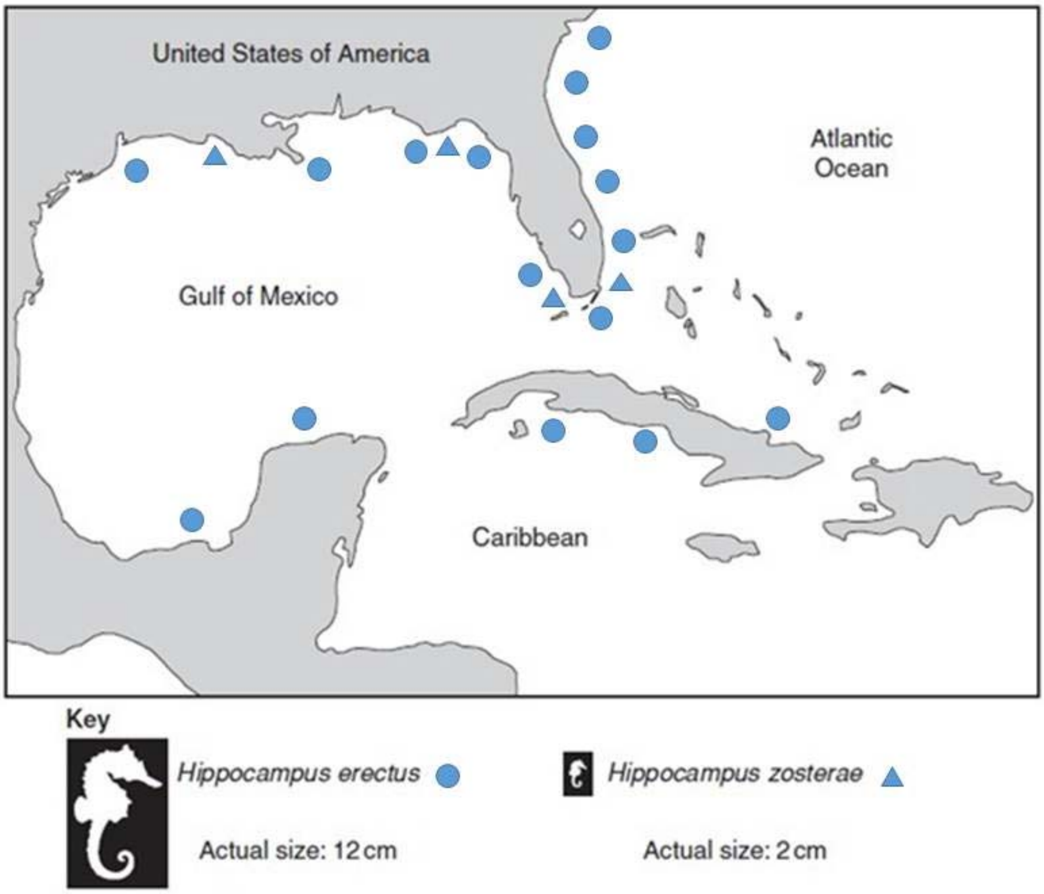

Fig. 8.1

Two different species of seahorse are found in the coastal region shown in Fig. 8.1. The ranges of these seahorse species overlap in many areas of these waters.

The two seahorse silhouettes are not drawn to scale.

(a) Using the information given, state the type of speciation that has occurred in the seahorses and explain your answer.

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(b) The type of natural selection that can produce the type of speciation that has occurred in seahorses is known as disruptive selection.

Explain how disruptive selection occurs in seahorse populations.

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(c) Fig. 8.2 shows the phylogenetic tree of three closely related species of seahorses based on nucleotide sequences, with ages estimated from fossils and biogeographical data.

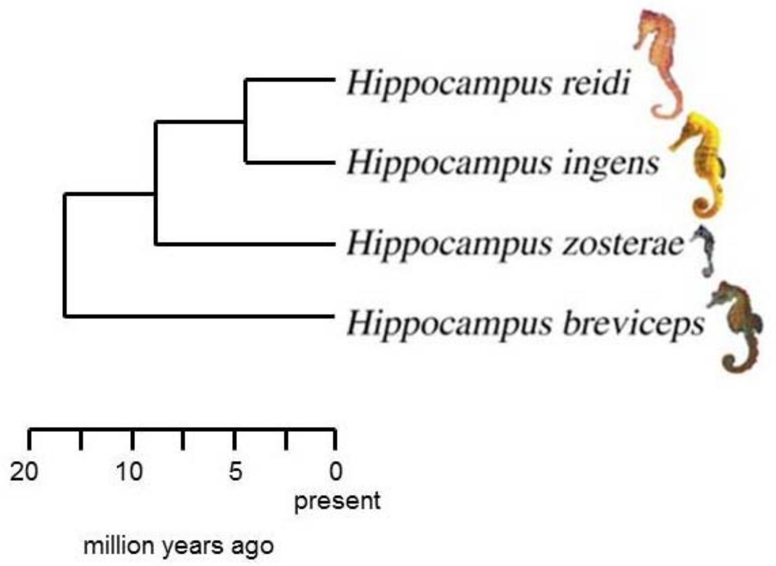

Fig. 8.2

Describe the advantages of using nucleotide sequences in reconstructing the phylogenetic relationships.

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[Total:9]

9 Fig. 9.1 shows the molecular structure of an antibody.

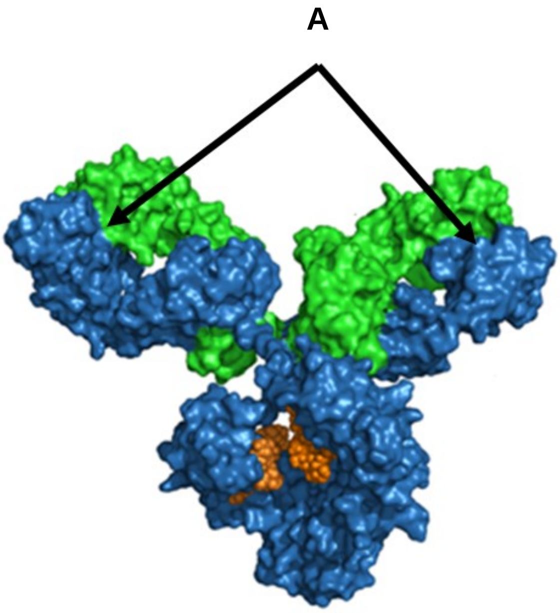

Fig. 9.1

(a) Explain how diversity at regions labelled **A** is generated in developing antibody-secreting cells.

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.... [2]

Fig. 9.2 shows the typical antibody concentration in the serum of a patient during a primary and a secondary immune response to the same antigen.

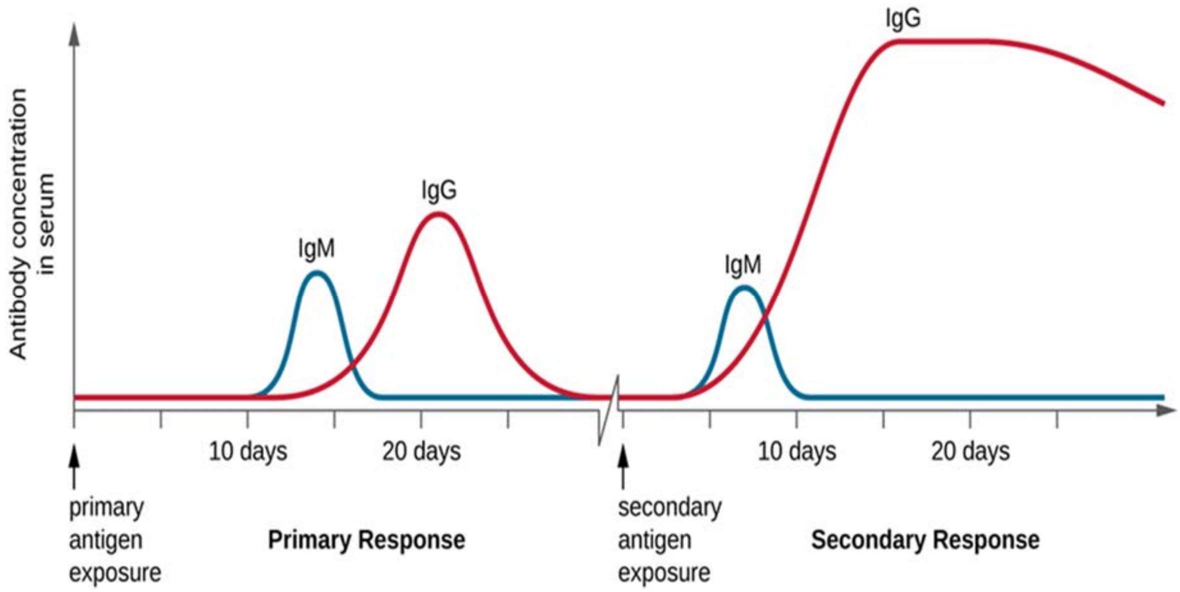

Fig. 9.2

(b) **(i)** State **two** significant differences between the primary and secondary immune responses.

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.... [2]

(ii) Explain the differences stated in **(bi)**.

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.... [3]

Fig. 9.3 shows the production of all blood cells from Cell **B**.

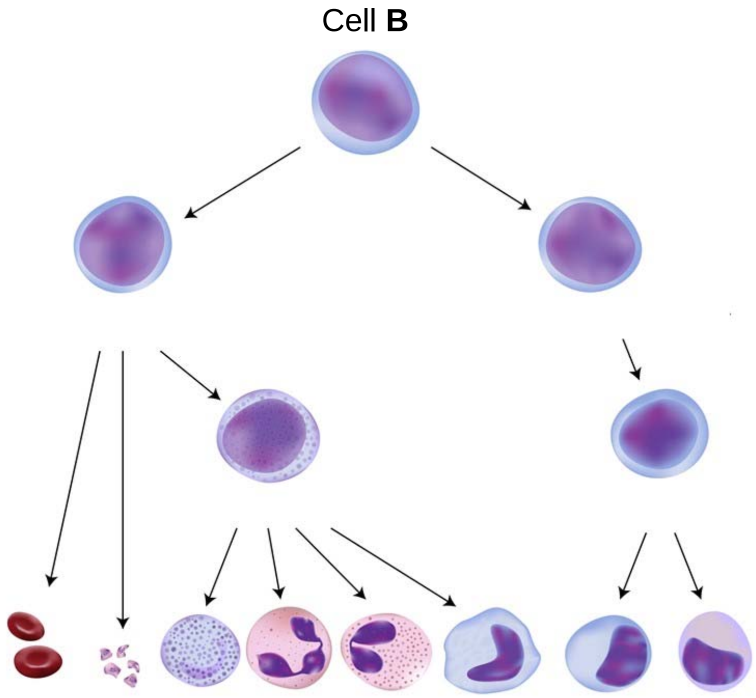

Fig. 9.3

(c) Identify Cell **B** and explain why, as shown in Fig. 9.3, it must have the characteristics of a stem cell.

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[Total: 11]

10 An analysis of ice cores from the Artic and Antarctic can provide information about the composition of the Earth's atmosphere over thousands of years.

Fig. 10.1 shows the concentrations of carbon dioxide and methane measured in ice cores, dated between 1000 and 2000AD.

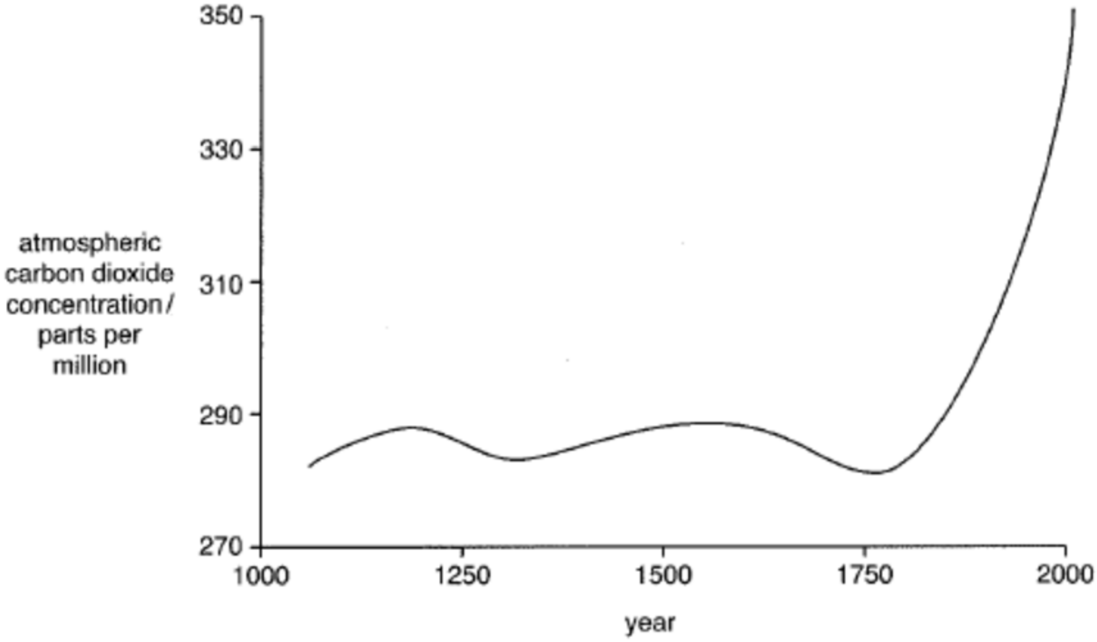

Fig.10.1

(a) Describe and explain the data in Fig. 10.1 from 1750AD onwards.

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.... [2]

(b) Explain how increasing concentrations of gases, such as carbon dioxide and methane, are thought to cause global warming.

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.... [2]

(c) As part of the Alliance of Small Island States (AOSIS), Singapore is considered a low-lying small island country. Like many other small island states, there are many fringing and patch coral reefs found around the smaller islands, south of Singapore mainland.

Describe how global warming can impact small island states like Singapore.

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[Total:8]